How well does public opinion perform against traditional finance benchmarks for interest rate hikes?

Marcus Guthe

July 2026

Abstract

Prediction markets have grown rapidly as a venue for forecasting macroeconomic events, including Federal Reserve interest rate decisions. This paper investigates whether a public prediction market – specifically Polymarket's FOMC meeting markets – can match or outperform a traditional, futures-derived benchmark for predicting Federal Reserve rate decisions. We first replicate the CME Group FedWatch Tool methodology from raw 30-Day Federal Funds futures (ZQ) data, meeting dates, and target rates, validating the replication against CME's own published probabilities (within $\pm 2\%$). We then compare this model against Polymarket's implied probabilities across 21 resolved FOMC meetings (2023–2026) along two dimensions: calibration, measured via the Brier score, and timing, measured via threshold-crossing lead time and lagged cross-correlation. The model achieves a lower Brier score in 15 of 21 meetings – a directionally consistent but not statistically robust advantage (sign test $p = 0.039$; paired t-test $p = 0.081$; Wilcoxon $p = 0.065$). The timing result is more robust: the model crosses confidence thresholds earlier than Polymarket in the large majority of meetings, with a median lead of several days at higher thresholds, and the cross-correlation between the two series peaks at a +2-day lag – consistent with Polymarket lagging the futures-implied signal. This gap is broadly consistent with prior evidence that deep, professional derivatives markets incorporate new information faster than retail-driven markets (e.g., Kawaller, Koch & Koch, 1987). Finally, we outline and backtest a simple threshold-and-Kelly-sizing strategy for exploiting this gap; while it shows a positive average PnL and no meeting with negative aggregate PnL in-sample, we show this is largely a mechanical consequence of the payoff structure rather than evidence of a reliable trading edge, since the backtest has not yet been tested against a meeting where the model's leading guess was wrong all the way to resolution.

1 Introduction and Background

Prediction markets have seen a large increase in the amount of money flowing into markets that forecast future events. One such market predicts what the Federal Reserve will decide regarding the federal funds rate at upcoming FOMC meetings. These markets usually have over $200 million wagered on the outcome of a single meeting. From a market this size, one might expect a wisdom-of-crowds effect to emerge and outperform traditional financial methods for estimating the future Fed rate. In this paper, however, we show that prediction markets are generally slower than traditional futures-based methods at pricing in the correct FOMC outcome. In short, this paper explores how well the general public can predict interest rate decisions compared to a traditional model based on Federal Funds futures, and whether any resulting mispricing can be exploited.

This paper explores this in three distinct stages:

1. Build a system that replicates the Chicago Mercantile Exchange (CME) methodology for predicting Federal Funds rate changes, using raw data such as meeting dates, target rates, and futures prices (ZQ).
2. Compare the resulting probabilities to Polymarket data over time.
3. Explore the potential mispricing and how one might use it to gain an edge in prediction markets.

2 Recreating CME's Federal Funds Rate Prediction Model

We replicate CME's FedWatch methodology from raw 30-Day Federal Funds futures (ZQ) data, meeting dates, and target rates. In short, each month's expected rate change is decomposed into a whole number of 25bp moves plus a remainder, which determines how probability is split between adjacent outcome levels for the nearest meeting; this distribution is then convolved forward through the meeting sequence. The full derivation, including the day-weighted decomposition for months containing a meeting, is given in Appendix A. Our replication matches CME's own published probabilities to within $\pm 2\%$.

3 Comparing Model Probabilities to Prediction Market Data

We now have a model, grounded in existing financial theory, for predicting future Federal Funds rate changes, which we can compare against the same prediction from a prediction market. The data used for comparison was limited to Polymarket's markets for "Federal decision in <Month>", to ensure consistent behavior across meetings. Since market volatility tends to be low far from the meeting date, which introduces a larger noise factor into any comparison, I limited the comparison to the 90 days immediately preceding each meeting to minimize noise. I did this using Kaufman's Efficiency Ratio, defined as the net movement over a window divided by the sum of each day's absolute movement within it, which produced Figure 1. The figure shows that signal is strongest

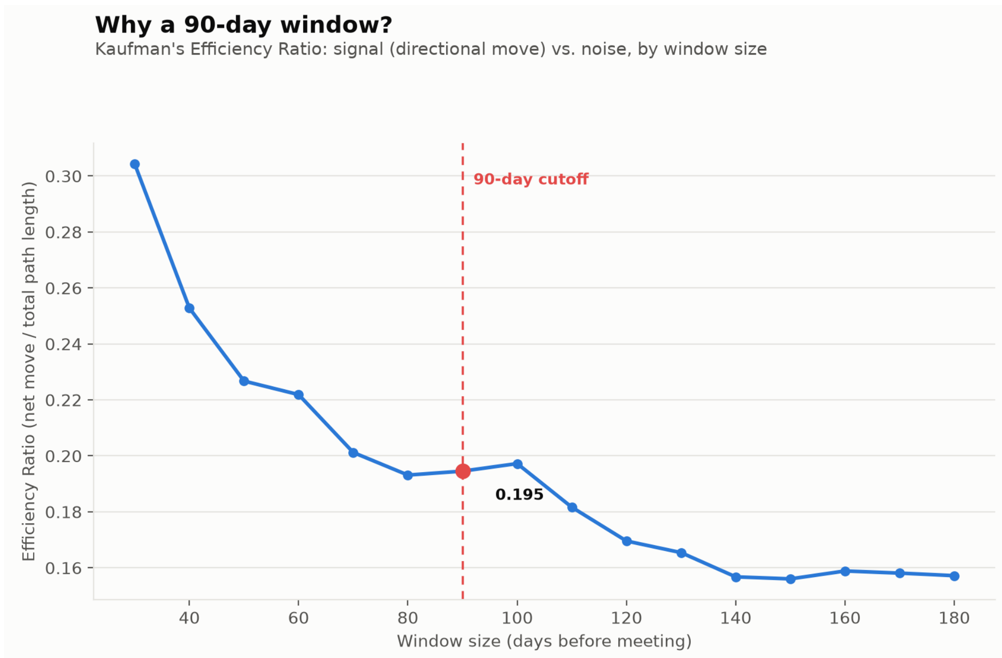

Figure 1: Kaufman's Efficiency Ratio

close to the meeting and falls off the further away we get. The curve is mostly monotonic, without a sharp elbow, so the exact choice of a 90-day window involves a degree of judgment: it had to balance the noise-to-signal ratio against spanning a long enough period to give our model a chance to differentiate itself from Polymarket, since the two series converge as the meeting approaches.

With this 90-day window established, the comparison was carried out in two stages: first, testing whether the model predicts the outcome with greater accuracy; second, testing whether the model predicts the correct outcome earlier in time.

3.1 Testing Model Accuracy

To compare the model's accuracy to the Polymarket data, I used the Brier score (BS), computed daily and averaged over the 90-day window for each meeting. Let i index a day within the window, let y_i be 1 if the event occurred and 0 otherwise, and let $\hat{p}_i$ be the estimated probability of the event (a given rate change) on day i. The Brier score is then

$$BS = \frac{1}{90} \sum_{i=1}^{90} (\hat{p}_i - y_i)^2. \quad (1)$$

The Brier score was calculated for each of the 25bp hikes and drops and then compared to the Brier score of the Polymarket data, calculated in the same way. Our results are shown in Table 1 below, as well as in the scatter plot in Figure 2. From the results, our model consistently achieves a lower (better)

Table 1: Comparison of Brier score: model vs. Polymarket (2023–2026)

	Model	**Polymarket**
Lower Brier score	15/21	6/21
Average Brier score	0.0347	0.0409
Median Brier score	0.0263	0.0275

Brier score: the average is approximately 15% better than Polymarket's, and the median is approximately 4% better. This is not a large improvement, and the average is strongly influenced by one data point: the unexpected rate decision in September 2024, shown in Figure 2.

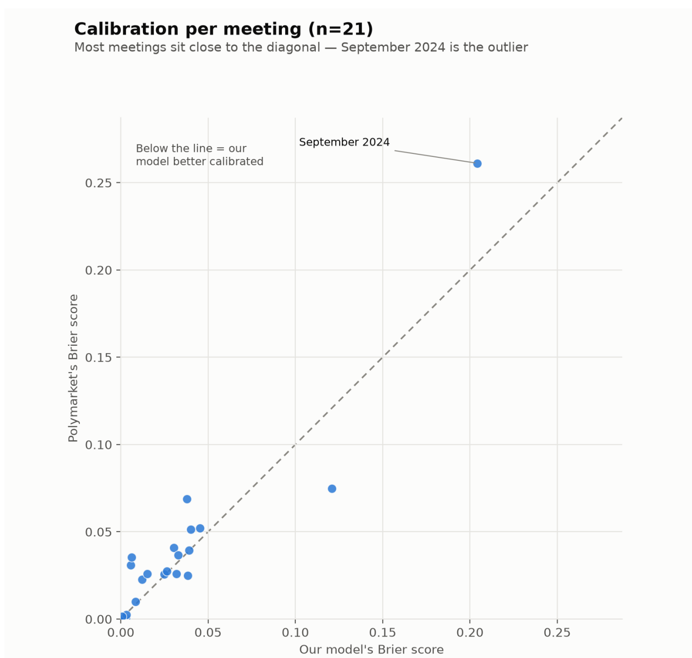

Figure 2: Comparison of Brier score: model vs. Polymarket (2023–2026)

The average score is nonetheless an interesting metric in its own right, since it shows that while the model performs better on "surprising" decisions, it is only marginally better for a typical meeting — as reflected in the median score.

The difference in Brier score is directionally consistent (15/21 meetings, sign test $p = 0.039$), but not statistically robust under more informative tests (t-test $p = 0.081$, Wilcoxon $p = 0.065$), and even these figures are likely optimistic given correlated observations. The result should be interpreted as an indication rather than a proven finding, especially given the low number of data points (only 21) and the likely correlation between them.

We therefore cannot conclude that our model is significantly better calibrated than Polymarket at predicting FOMC meeting outcomes. This does not, however, rule out a lead-time advantage: our analysis in fact finds that while the model is not significantly more accurate on average, it does tend to reach a confident, correct conclusion earlier in time than Polymarket. This is explored in the following subsection.

3.2 Testing the Model's Information Time-Gap Advantage

While the above results do not show that our model is significantly more accurate, it might still produce a lead-window. Looking at the graphs in Figure 3, we might qualitatively say that our model tends to reach the right answer earlier.

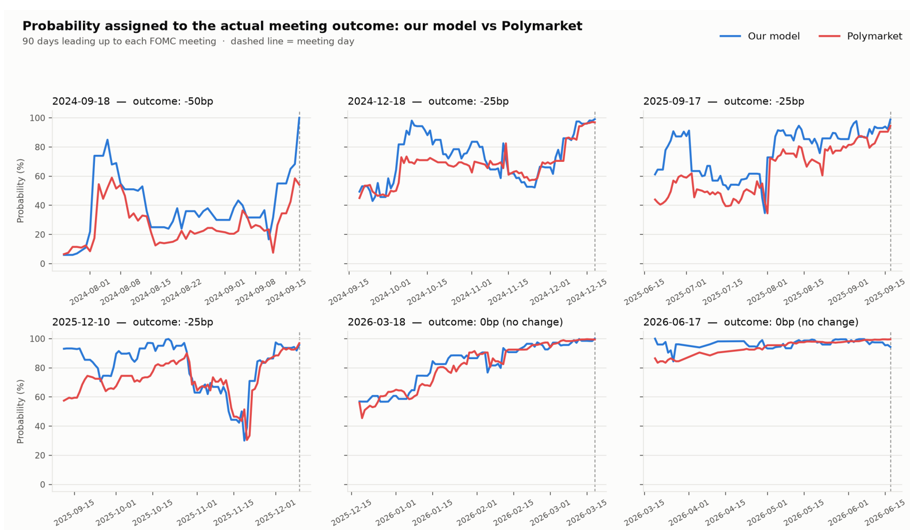

Figure 3: Model (blue) vs. Polymarket (red) rate prediction certainty

To test this more rigorously, I used two different methods.

3.3 Threshold Passing for Different Levels

For every resolved meeting, we find the first day on which the model's and Polymarket's probabilities for the correct outcome cross an established threshold. If the model crosses this threshold before Polymarket, we say it has a *lead-window*. I tried this for five threshold values between 55% and 90%, resulting in Table 2.

Table 2: Who crossed the threshold first: model vs. Polymarket

Threshold	**n**	**Model first**	**Poly. first**	**Tied**	**Mean lead**	**Median lead**
55%	21	8	1	12	2.0 days	0.0 days
60%	20	10	2	8	4.2 days	0.5 days
70%	20	11	1	8	10.2 days	1.0 days
80%	20	13	1	4	12.3 days	3.5 days
90%	20	14	5	1	19.2 days	9.0 days

(n drops from 21 to 20 at thresholds $\geq 60\%$ because Polymarket's price for the actual outcome in September 2024 never reached 60% within the 90-day window; a lead cannot be calculated if one side never crosses the threshold. This is an expected consequence of the threshold choice, not a data problem.)

Across all five thresholds, our model had a lead-window in substantially more meetings than Polymarket did. Note that the median lead is a more robust measure than the mean, since the mean is heavily influenced by a few meetings. One critique of this method is that it introduces a substantial look-ahead bias, since we already know the eventual outcome. This is not a problem here, however, since we are measuring the convergence rate given a known answer, rather than how trustworthy a live signal would be in real time — something to keep in mind when implementing this as a live strategy.

3.4 Cross-Correlation

Threshold passing is sensitive to where one places the threshold, and to outliers. As a separate control, I centered both time series per meeting by subtracting each meeting's average, to isolate relative movement over time, and then calculated the correlation between the model's probability today and Polymarket's probability at different lags. Table 3 shows that the correlation peaks at a

Table 3: Cross-correlation between model and Polymarket series at varying lags

Lag	−8 days	−4 days	0 days	+2 **days**	+4 days	+8 days
Correlation	0.505	0.641	0.753	**0.790**	0.742	0.597

+2-day lag, indicating that Polymarket's probability two days into the future matches our model's current probability more closely than Polymarket's own current probability does. This is a characteristic typical of lagging time series, and suggests our model tends to be faster at pricing in new information.

3.5 Statistical Significance of the Time-Lag Findings

I ran three formal tests on the paired differences (our model minus Polymarket) for the lead-time result: a paired t-test, a Wilcoxon signed-rank test, and a sign test. The results are shown in Table 4.

Table 4: Formal significance tests on paired differences (our model minus Polymarket)

Test	**Testing**	**Lead time, 80% threshold**
Paired t-test	**Mean** of the differences	$t = 3.00$, $\mathbf{p = 0.0037}$
Wilcoxon signed-rank	**Median** of the differences	$W = 16$, $\mathbf{p = 0.0071}$
Sign test (binomial)	Share of meetings we won	13/16, $\mathbf{p = 0.011}$

(All p-values one-sided, against the hypothesis that our model is better/earlier. The lead-time sign test excludes 4 meetings with a tied outcome at the 80% threshold, hence $n = 16$ *rather than 20.)*

Our model's time advantage is more statistically robust than its ability to predict a rate hike or cut. As before, however, we still have the same issue of a low number of data points, and the data being highly correlated, so the above p-values are likely optimistic.

One can argue that this finding makes sense: Federal Funds futures trade on a deep, professional market where institutional traders and market makers adjust prices immediately on new information, whereas Polymarket is a more retail-driven market where participants are less informed and prices move more slowly on new information. This has some support in the existing literature — Kawaller, Koch & Koch (1987), "The Temporal Price Relationship between S&P 500 Futures and the S&P 500 Index," *Journal of Finance*, showed that S&P 500 futures often lead the S&P 500 index by 20–45 minutes intraday, consistent with the idea that a deep, low-friction derivatives market moves more quickly on new information than a slower, thinner market.

4 Implementation of Our Findings as a Trading Strategy

While our findings are not necessarily statistically robust, I wanted to explore how one might trade on this time-gap discrepancy. In this section I outline one such strategy. It should be noted that the calculations below do not account for slippage or trading fees.

- **Instrument:** individual outcome levels on Polymarket's canonical FOMC meeting markets, within the 90-day window.
- **Entry:** the model's probability p above a configurable threshold T, *and* $p > P$ (Polymarket's price) — a positive edge, a condition required for Kelly-based position sizing to be meaningful.

- **Exit:** when another level takes over as the model's leading guess, or when the meeting is resolved.
- **Sizing:** Kelly fraction $f^* = (p - P)/(1 - P)$, in practice fractional ($\frac{1}{4}$–$\frac{1}{2}\times$) with a hard cap per position — full Kelly averaged 40.6% of bankroll per trade and up to 100% in individual cases, which is too aggressive to use unmodified.

Table 5: Backtest performance by entry threshold

Threshold	Trades	Meetings w/ negative PnL	Mean PnL/meeting	Hit rate
50%	64	0/21	+1.10 kr	67.2%
60%	45	0/21	+0.94 kr	66.7%
70%	40	0/21	+0.73 kr	62.5%

A single meeting (September 2024, a genuinely more surprising decision than the rest of the period) contributed roughly 11% of the total backtest PnL at $T = 60\%$ — small enough that the result should not be interpreted as depending on a single outlier.

One might ask why no single meeting ever ends up negative. Broken down by exit type ($T = 60\%$; see the **Exit** definition above), we can see in Table 6:

Table 6: Backtest PnL by exit type ($T = 60\%$)

Exit type	Number of trades	Mean PnL	Min	Max
(a) Superseded	24	+0.09 kr	−0.15 kr	+0.86 kr
(b) Resolved	21	+0.84 kr	+0.01 kr	+1.99 kr

The superseded positions have real losses (down to −15%), which is where the strategy's noise actually shows up. But all 21 positions held all the way to resolution won, including September 2024. Because a meeting's total PnL is dominated by the binary outcome (0 or 1 kr) of the final, resolving trade, it only takes that one component always winning for no single meeting to end up negative even when individual superseded sub-positions run at a loss.

One weakness this reveals is that the backtest has never been tested against a meeting where the model stuck with the wrong level all the way to the end.

5 Conclusion

This paper set out to answer two related questions: can a simple, publicly replicable model based on Federal Funds futures data match or beat Polymarket's crowd-sourced probabilities for FOMC rate decisions, and if a gap exists, can it be exploited?

On the first question, the results are modest and should be read cautiously. Our replication of the CME FedWatch methodology tracks CME's own published probabilities closely and produces a lower Brier score than Polymarket in 15 of 21 resolved meetings. However, with only 21 meetings — and with meeting-to-meeting errors plausibly correlated through shared macro regimes — none of the more informative tests (paired t-test, Wilcoxon) reach conventional significance, and even the sign test's p-value is likely optimistic once that correlation is accounted for. The honest conclusion is that our model is directionally, but not demonstrably, better calibrated than Polymarket.

The timing result is more convincing. Across five different confidence thresholds, the futures-derived model consistently reaches a high-confidence, correct answer earlier than Polymarket, and this lead survives all three significance tests at conventional levels. The lagged cross-correlation analysis, which does not depend on an arbitrarily chosen threshold, tells the same story: Polymarket's probability two days into the future resembles our model's current probability more closely than Polymarket's own current probability does. Taken together, this is consistent with the broader finding in the market-microstructure literature that deep, professional derivatives markets impound new information faster than thinner, retail-driven venues (e.g., Kawaller, Koch & Koch, 1987, for S&P 500 futures versus the cash index) — though this analogy should not be overstated, since Federal Funds futures and a binary event-prediction market differ in several structural respects (contract design, liquidity profile, and participant base) that this paper has not directly controlled for.

The illustrative trading strategy in Section 4 should not be read as evidence of a genuine, exploitable edge. Its clean backtest record — no resolved meeting with negative aggregate PnL — is substantially explained by the mechanics of the payoff itself: because the final, resolving position pays out binarily and has so far always been on the correct side, a single bad "superseded" sub-position is never large enough to flip a meeting's total PnL negative. The more important finding is diagnostic, not commercial: the backtest has never faced a meeting where the model's leading guess was wrong all the way to resolution, because the 2023–2026 sample happened to contain very few genuine surprises. That is precisely the scenario in which this strategy would be expected to lose money, and until it has been tested against one — either by waiting for such a meeting to occur out-of-sample, or by stress-testing against a historical period with more Fed surprises — no claim about its live viability is warranted. Slippage, fees, and Polymarket's comparatively thin liquidity far from the meeting date would further erode the backtested returns.

A CME FedWatch: full derivation

As outlined on CME's methodology page, the model rests on a few key assumptions. In summary:

- Rate hikes and cuts are assumed to occur in uniform 25bp increments, with the Effective Federal Funds Rate (EFFR) reacting proportionally to the size of the move.
- The probability of a hike (or cut) is obtained by summing the probabilities of all target-rate levels above (or below) the current target.
- Each month's Federal Funds futures contract price is taken to reflect the market's expectation of that month's average daily EFFR.
- FOMC meetings follow a largely regular published schedule, with one month per quarter typically having no meeting.
- The projected end-of-month EFFR in one month is assumed to equal the start-of-month EFFR in the next.

Since these are simplifying assumptions, the resulting probabilities should be read as estimates that can be affected if any of the underlying assumptions are violated.

With these assumptions in place, we also need to define $EFFR_T(\text{end})$ and $EFFR_T(\text{start})$, since both are required to calculate the expected number of rate hikes or cuts. They are defined as follows:

- If month T does *not* contain a FOMC meeting, we can use the continuity of EFFR across month boundaries to state that

  $$EFFR_{T-1}(\text{end}) = EFFR_T(\text{avg}) = EFFR_{T+1}(\text{start}), \tag{2}$$

  where $EFFR_T(\text{avg}) = 100 - ZQ_T$, since for a month without a meeting that month's futures contract price represents the average EFFR for the entire month.

- If, however, month T *does* contain a FOMC meeting — call such a month T' — the relation in (2) no longer determines $EFFR_{T'}(\text{start})$ and $EFFR_{T'}(\text{end})$ directly, since the rate can change partway through the month. Let K be the number of days in the month up to and including the meeting date, and M the number of days after the meeting. $EFFR_{T'}(\text{start})$ and $EFFR_{T'}(\text{end})$ are then calculated as follows, depending on which of the two is already known from an adjacent non-meeting month via (2):

  $$EFFR_{T'}(\text{start}) = \frac{EFFR_{T'}(\text{avg}) - \frac{M}{M+K} \cdot EFFR_{T'}(\text{end})}{\frac{K}{M+K}} \tag{3}$$

  $$EFFR_{T'}(\text{end}) = \frac{EFFR_{T'}(\text{avg}) - \frac{K}{M+K} \cdot EFFR_{T'}(\text{start})}{\frac{M}{M+K}} \tag{4}$$

With our assumptions and formulas in place, we can now give a more thorough walk-through of the method.

A.1 Step 1

Start by finding the nearest month without a FOMC meeting and use (2) to fill in $EFFR_T(\text{start})$ and $EFFR_T(\text{end})$ for the months immediately preceding and following it.

A.2 Step 2

For any neighboring month that does contain a meeting, calculate $EFFR_{T'}(\text{start})$ and $EFFR_{T'}(\text{end})$ using (3) or (4), since by assumption a month adjacent to a non-meeting month must itself contain a meeting. Repeat this process until every month has values for $EFFR_T(\text{start})$, $EFFR_T(\text{end})$, and $EFFR_T(\text{avg})$.

A.3 Step 3

Calculate the expected monthly change in EFFR for the nearest upcoming meeting month:

$$\text{Expected } EFFR \text{ change} = EFFR_{T'}(\text{end}) - EFFR_{T'}(\text{start}) \tag{5}$$

A.4 Step 4

Convert this into the equivalent number of 25bp hikes: Expected $EFFR$ change/0.25.

A.5 Step 5

Split this number into its integer part, N, and remainder, r. The remainder r then determines how probability is split between the two adjacent 25bp outcomes for the nearest upcoming meeting.

A.6 Step 6

Steps 1–5 give the probability distribution for a single meeting. To calculate the probability for future meetings, repeat steps 1–5 for the next meeting, then convolve the resulting distribution forward through the entire meeting sequence to obtain the probability of a rate change at each future meeting.

Using the steps above, I compared my model bin-by-bin to CME's own published data and found that it matched within ±2%.